Work in progress: A Lean-verified conditional reduction of periodic Navier–Stokes regularity on $\mathbb{T}^3$ to a single integrability hypothesis

(conditional results; draft)

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Scope. This paper reports a *conditional* result. It does **not** solve the Clay Millennium Problem, and it does not prove global regularity of the 3D Navier–Stokes equations. The single remaining hypothesis, `PeriodicBKMIntegrability`, is a named Prop that is **not proved**. It is equivalent in difficulty to Clay Statement (B).

Abstract

In the Lean 4/Mathlib repository `BenFrohman/periodic-ns-classical-continuation` (commit `4ab624e`), we prove two results with no hypothesis Props:

- H^3 Beale–Kato–Majda control on $\mathbb{T}^3$, `periodicBKMHsControl_three`;
- the periodic BKM blow-up criterion, `periodicBKMCriterion`.

We deduce the exact periodic Clay Statement (B), `Clay_B_exact`, *conditionally* on the single named Prop `PeriodicBKMIntegrability` (`clay_B_exact_of_integrability`). The axioms used are only `propext`, `Classical.choice` and `Quot.sound`.

1 The periodic chain on $\mathbb{T}^3$

Throughout, $\nu > 0$, u_0 is smooth, periodic and divergence-free, and (u, p) is the maximal smooth solution with existence time T^*. The definitions `Clay_B_exact`, `PeriodicBKMCriterion` and `PeriodicBKMIntegrability` come from the dependency `fefferman-clay-navier-stokes`.

Hypothesis 1 (`PeriodicBKMIntegrability` ν; **open**)**.** For every maximal solution and every T, $t \mapsto \|\omega(t)\|_\infty$ is integrable on $[0, T]$.

Theorem 1 (`periodicBKMHsControl_three`†; unconditional)**.** *For every ν, `PeriodicBKMHsControl` ν 3 holds. That is, if $T^* = T < \infty$ and $\int_0^T \|\omega\|_\infty < \infty$, then $\sup_{[0,T)} \|u(t)\|_{H^3} < \infty$.*

Proof route (all formalized). The H^3 energy identity reduces the claim to a commutator bound and a low-frequency vorticity bound.

- The $s = 3$ commutator bound is $-2\sum_i \int (1-\Delta/4\pi^2)^3 v_i\,((v\cdot\nabla)v)_i \le C\|\nabla v\|_\infty \|v\|_{H^3}^2$ (`periodicPhysicalComm` `CommutatorAssembly.lean`).

- The low-frequency bound is $\|V_R \nabla v\|_\infty \le K(1+\log R)\|\omega\|_\infty$ for $R \ge 2$ (`periodicLowFreqVorticityBound`†, `LowFreqAssembly.lean`). It is proved through the Riesz form of Biot–Savart, a smooth dyadic decomposition with uniformly L^1-bounded block kernels, and a uniform L^1 bound for the de la Vallée Poussin kernel.

Grönwall's inequality with the log splitting then closes the bound.

Theorem 2 (`periodicBKMCriterion`†; unconditional)**.** *For every ν, `PeriodicBKMCriterion` ν holds. If $\nu > 0$ and $T^* = T < \infty$, then $\int_0^T \|\omega\|_\infty = \infty$.*

Theorem 3 (`clay_B_exact_of_integrability`†; conditional)**.** *`PeriodicBKMIntegrability` $\nu \Rightarrow$ `Clay_B_exact` ν.*

Remark 1 (Equivalence in difficulty)**.** The converse direction also holds. If Statement (B) gives a global smooth solution, then by uniqueness the maximal solution is global and smooth, so $\|\omega\|_\infty$ is continuous and integrable on every $[0, T]$. Hence `PeriodicBKMIntegrability` is *equivalent* to the periodic Clay Statement (B); it is not a weaker or "technical" leftover. This converse, from (B) itself, is argued informally here. What *is* proved in Lean (`periodicBKMIntegrability_iff_globalLifespan`, `IntegrabilityAttack.lean`) is that `PeriodicBKMIntegrability` is equivalent to `PeriodicGlobalLifespan`, i.e. $T^* = \infty$ for all smooth data. Its proof uses `integrableOn_vorticity_of_lt_tstar`: $\int_0^T \|\omega\|_\infty < \infty$ whenever $T < T^*$. this paper is therefore a verified reduction, not a proof of regularity.

2 Stated vs. proved

Statement	Status
H^3 BKM control, `periodicBKMHsControl_three`	proved unconditionally
Commutator bound ($s = 3$), `periodicPhysicalCommutatorBound`	proved unconditionally
Low-frequency vorticity bound, `periodicLowFreqVorticityBound`	proved unconditionally
Periodic BKM criterion, `periodicBKMCriterion`	proved unconditionally
Clay Statement (B), via `clay_B_exact_of_integrability`	proved conditionally on `PeriodicBKMIntegrability`
`PeriodicBKMIntegrability`	not proved (open; equivalent to Clay (B))
Converse: (B) $\Rightarrow$ `PeriodicBKMIntegrability`	not proved in Lean (informal argument only)
`periodicBKMIntegrability_iff_globalLifespan`†	proved unconditionally
`integrableOn_vorticity_of_lt_tstar`† (below T^*)	proved unconditionally
Small data, `integrability_of_smallDataGlobal`†	proved from named `PeriodicSmallDataGlobal` (not formalized)
Prodi–Serrin, `integrability_of_prodiSerrin`†	proved from named `PeriodicProdiSerrinCriterion` (not formalized)

$\kappa = 0$ route, `periodicBKMIntegrability_of_vorticityAPriori`	proved from `PeriodicVorticityAPrioriBound` (APri)†
Clay problem / NS global regularity	not proved, not claimed

A Theorem-to-Lean map

Names marked † were coined during formalization and are not the authors terminology; they may be renamed later.

Lean name	File	Repo @ commit
`periodicBKMHsControl_three`†	`PeriodicNSContinuation/BKMFinal.lean`	pnscc @ `4ab624e`
`periodicBKMCriterion`†	`PeriodicNSContinuation/BKMFinal.lean`	pnscc @ `4ab624e`
`clay_B_exact_of_integrability`†	`PeriodicNSContinuation/BKMFinal.lean`	pnscc @ `4ab624e`
`periodicBKMHsControl_three_of_lowFreq`†	`PeriodicNSContinuation/CommutatorAssembly.lean`	pnscc @ `4ab624e`
`periodicPhysicalCommutatorBound`†	`PeriodicNSContinuation/CommutatorAssembly.lean`	pnscc @ `4ab624e`
`periodicLowFreqVorticityBound`†	`PeriodicNSContinuation/LowFreqAssembly.lean`	pnscc @ `4ab624e`
`periodicBKMCriterion_of_hsControl`†	`PeriodicNSContinuation/Continuation.lean`	pnscc @ `4ab624e`
`PeriodicBKMHsControl`† (def)	`PeriodicNSContinuation/Continuation.lean`	pnscc @ `4ab624e`
`clay_B_exact_of_bkm`†	`PeriodicNSContinuation/LocalExistence.lean`	pnscc @ `4ab624e`
`Clay_B_exact` (def)	`FeffermanClayNS/FeffermanStatement.lean`	dependency
`PeriodicBKMCriterion`, `PeriodicBKMIntegrability` (defs)	`FeffermanClayNS/HighSetDepletionAndBKM.lean`	dependency

pnscc = `BenFrohman/periodic-ns-classical-continuation` (private). "dependency" = `fefferman-clay-na` as pinned in the lake manifest.

B Axiom audit

At commit `4ab624e`, `lake build` succeeds with 0 `sorry`, and `lake env lean AxiomAudit.lean` prints "AxiomAudit PASS: only propext, Classical.choice, Quot.sound". Every theorem above is in the audit.

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